$F = k \cdot \frac{Q_1 \cdot Q_2}{r^2}$

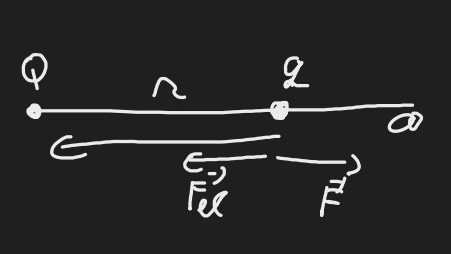
$E = k \cdot \frac{Q_1}{r^2}$

$\boxed{F = q \cdot E}$

$V = \frac{kQ}{r} + c$

$E = -grad(V)$

$U = \frac{L}{q}$
$\downarrow$
$\frac{kQ}{r}$



$C = \frac{\varepsilon_0 \cdot S}{d}$

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$E = 50000 \frac{N}{C}$

$d = 2\ mm = 2 \cdot 10^{-3}\ m$

$q_p = 1{,}6 \cdot 10^{-19} C$

$m_p = 1{,}67 \cdot 10^{-27} kg$

$L = U$

$E - d$

$E_p = L = \underbrace{F \cdot d} = q_p \cdot E \cdot d = \ldots$

$E_p = E_c \Rightarrow q_p \cdot E \cdot d = \frac{m_p \cdot v^2}{2} \Rightarrow \boxed{v = \sqrt{\frac{2 \cdot q_p \cdot E\ d}{m_p}}}$

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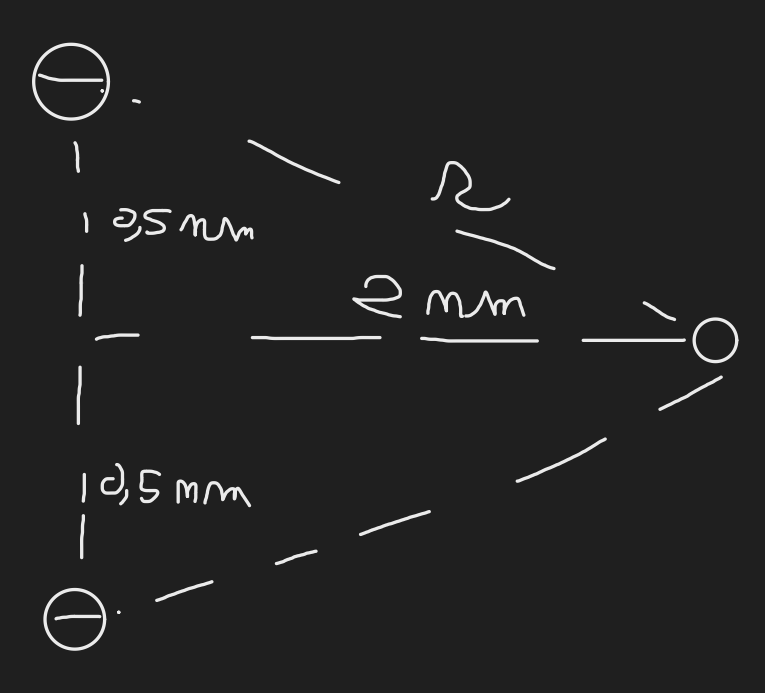

$U = \frac{L}{q} \Rightarrow L = q \cdot U$

$U = \frac{k q_e}{r}$

$\Rightarrow L = k \frac{q \cdot q_e}{r}$

$\boxed{E_p = 2L}$

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$U = \frac{L}{q_e} \Rightarrow L = q_e \cdot U$

$E_c = U \cdot q_e \Rightarrow \frac{m_e \cdot v^2}{2} = U \cdot q_e$

$\Rightarrow v = \sqrt{\frac{2U \cdot q_e}{m_e}}$

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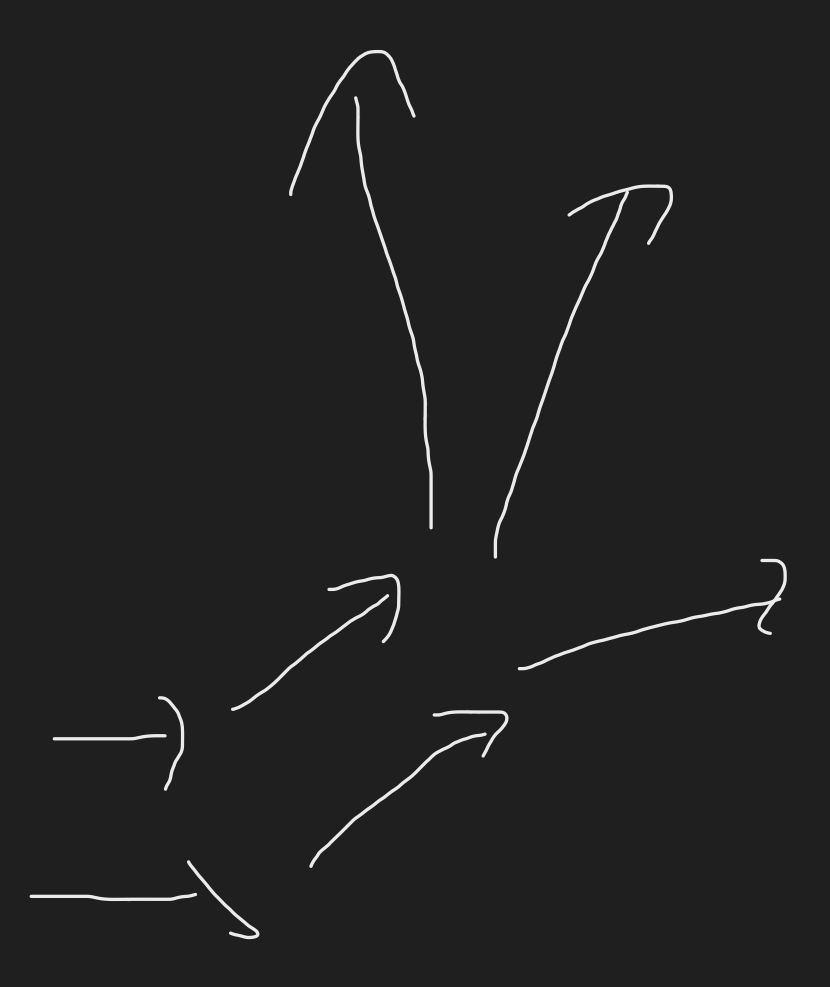
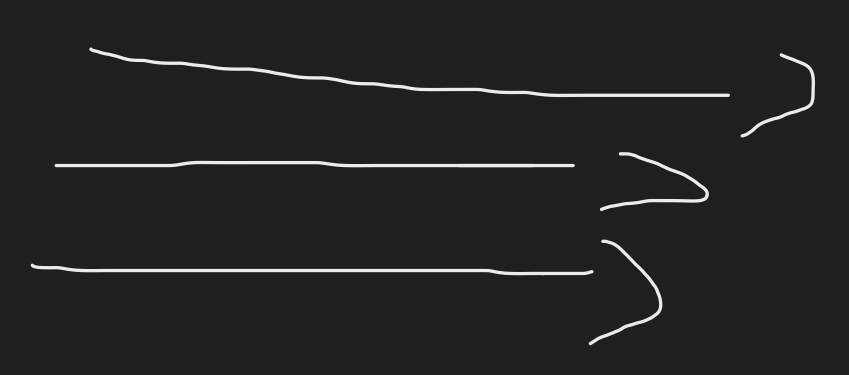
$V = (150x^2 - 200y^2) V$

$E = ?$ $\quad (x, y) = (2\ m, 2\ m)$

$\varphi = ?$





$E = -grad(V)$

$grad(V) = \frac{\partial V}{\partial x} \cdot \vec{i} + \frac{\partial V}{\partial y} \cdot \vec{j}$

$\frac{\partial V}{\partial x} = 300x$

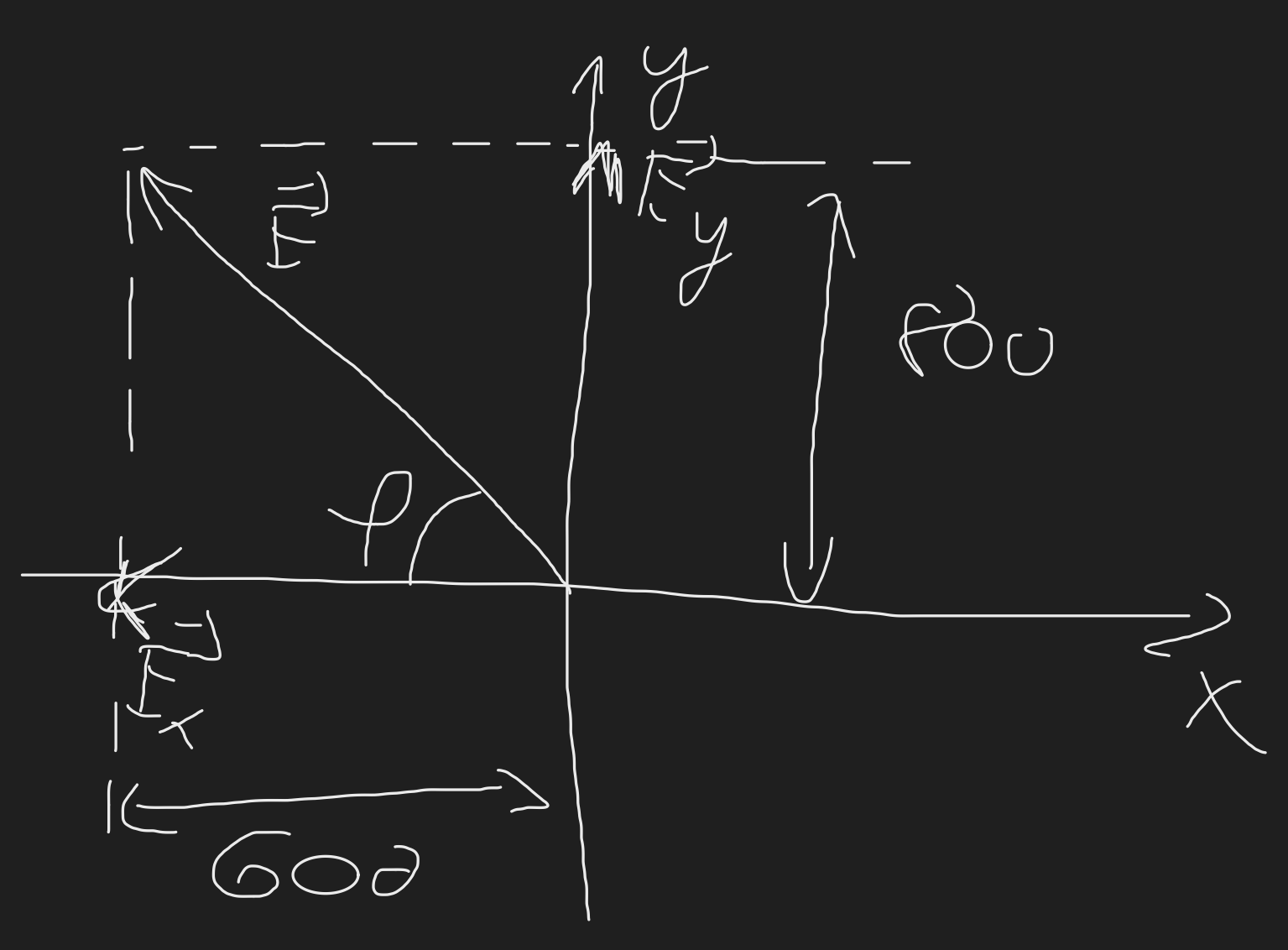
$\frac{\partial V}{\partial y} = -400y$

$\Rightarrow E = -\left(300x \cdot \vec{i} - 400y \cdot \vec{j}\right) \frac{N}{C} \left(\frac{V}{m}\right)$

$\vec{E} = -600\vec{i} + 800\vec{j}$

$|\vec{E}| = \sqrt{(-600)^2 + 800^2} = \sqrt{360000 + 640000}\ \frac{N}{C}$

$= \sqrt{1000\,000}\ \frac{N}{C} = 1000\ \frac{N}{C}$



$tg\ \varphi = \frac{800}{600}$

$\varphi = arctg\left(\frac{800}{600}\right)$